

Formal Characterization of an Airline Ticket Price Estimation System Using CatBoost

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Abstract

This text formalizes the machine learning model used in a system for estimating expected airline ticket prices. The problem is treated as supervised regression on tabular data, with two generalization regimes: *warm-start*, for historically represented routes, and *cold-start*, for evaluation on routes entirely absent from the corresponding training set. The estimator is based on CatBoost, and its purpose is to approximate the conditional expected value of price given a set of structural flight attributes and search-context variables. The model is not formulated as a temporal forecasting system for future prices.

1 Statistical objective

Consider an observation i associated with a flight searched at a given point in time. Let

$$Y_i \in \mathbb{R}_+$$

denote the observed ticket price in U.S. dollars, and let

$$X_i = (X_i^{\text{cat}}, X_i^{\text{num}})$$

be the vector of covariates used by the model.

The categorical component is

$$X_i^{\text{cat}} = (\text{flight_from}, \text{flight_to}, \text{route}, \text{company}),$$

while the numerical and binary component is

$$X_i^{\text{num}} = \begin{pmatrix} \text{departure_minutes} \\ \text{arrival_minutes} \\ \text{duration_minutes} \\ \text{stops} \\ \text{self_transfer} \\ \text{connection_count} \\ \text{days_until_departure} \\ \text{departure_weekday} \\ \text{search_hour} \end{pmatrix}.$$

The response variable is

$$Y_i = \text{price_usd}_i.$$

The goal is to construct an estimator

$$f : \mathcal{X} \longrightarrow \mathbb{R}_+$$

such that

$$\hat{Y}_i = f(X_i)$$

approximates the expected price conditional on the observed characteristics.
The central statistical interpretation is

$$\boxed{f(x) \approx \mathbb{E}[Y \mid X = x]}$$

that is, the model aims to estimate the conditional expectation of ticket price given the flight attributes and the search context.

Remark 1. *This formulation is distinct from forecasting a temporal price trajectory. The model estimates a price reference conditional on the current observed state, not the future value of the ticket.*

2 Regression model

The function f is approximated by a **CatBoost Regressor**, which belongs to the family of tree-based *gradient boosting* methods.

An abstract representation of the predictor is

$$\hat{f}_M(x) = f_0(x) + \sum_{m=1}^M \eta h_m(x),$$

where:

- f_0 is the initial predictor;
- h_m is the tree added at iteration m ;
- M is the total number of trees;
- $\eta > 0$ is the learning rate.

In abstract form, training seeks a function

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \sum_{i \in \mathcal{D}_{\text{train}}} L(Y_i, f(X_i)),$$

where L denotes the regression loss function used during training and $\mathcal{F}$ is the class of models under consideration.

CatBoost is well suited to the structure of the problem because the data are tabular and a relevant portion of the covariates is categorical, including airports, airlines, and routes.

3 Model-selection criterion

Candidate configurations are compared using the mean absolute error on the validation partition:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|.$$

The selection procedure follows the structure

$\mathcal{D}_{\text{train}} \longrightarrow \text{fit candidate models} \longrightarrow \mathcal{D}_{\text{val}} \longrightarrow \text{model selection} \longrightarrow \mathcal{D}_{\text{test}} \longrightarrow \text{final evaluation}.$

The test partition is not used to choose the winning configuration.

4 Two generalization regimes

The system maintains two regression models:

$$f_{\text{warm}} \quad \text{and} \quad f_{\text{cold}}.$$

They do not represent two different algorithms, but rather two training and evaluation regimes designed for different generalization problems.

4.1 Warm-start

Under the *warm* regime, the route belongs to the universe of routes historically represented in the dataset used to train the corresponding model.

Let

$$\mathcal{R}_{\text{warm}}$$

denote this set. In production, the condition

$$r \in \mathcal{R}_{\text{warm}}$$

determines whether the warm model is used.

In version V6,

$$|\mathcal{R}_{\text{warm}}| = 577.$$

The problem evaluated in this regime is generalization to new observations from historically known routes.

The selected warm configuration was

$$\text{warm_depth8_baseline}.$$

Final test-set metrics were:

Metric	Result
MAE	65.96 USD
RMSE	104.93 USD
R^2	0.8018
MAPE	25.76%

4.2 Cold-start

The *cold* regime was designed to measure generalization to routes not observed during training of the corresponding model.

Let

$$\mathcal{R}_{\text{train}}, \quad \mathcal{R}_{\text{val}}, \quad \mathcal{R}_{\text{test}}$$

denote the route sets associated with the training, validation, and test partitions.

The evaluation strategy imposes

$$\mathcal{R}_{\text{train}} \cap \mathcal{R}_{\text{val}} = \emptyset,$$

and

$$\mathcal{R}_{\text{train}} \cap \mathcal{R}_{\text{test}} = \emptyset.$$

Version V6 used 13 596 observations for training, 2 825 for validation, and 3 223 for testing. In terms of routes, 515 were used for training, 110 for validation, and 111 for testing. Two configurations of interest produced nearly identical validation performance:

$$\text{MAE}(\text{cold_route_depth8}) = 80.6460,$$

$$\text{MAE}(\text{cold_no_route_depth5_l2_15}) = 80.6521.$$

Under the adopted rule of selecting the model with the lowest validation MAE, the chosen configuration was

cold_route_depth8.

Final test-set metrics were:

Metric	Result
MAE	66.67 USD
RMSE	96.86 USD
R^2	0.7570
MAPE	19.98%

Remark 2. *The presence of the route variable in the cold model does not invalidate the cold-start experiment. The key point is that the route categories evaluated in validation and testing do not appear in the training set of that model.*

5 Production routing rule

The choice between the warm and cold models is made by a deterministic rule based on the set of routes known by the warm model.

Define

$$g(r) = \begin{cases} \text{warm}, & r \in \mathcal{R}_{\text{warm}}, \\ \text{cold}, & r \notin \mathcal{R}_{\text{warm}}. \end{cases}$$

The served prediction is therefore

$$\hat{Y}(X, r) = \begin{cases} f_{\text{warm}}(X), & r \in \mathcal{R}_{\text{warm}}, \\ f_{\text{cold}}(X), & r \notin \mathcal{R}_{\text{warm}}. \end{cases}$$

The set $\mathcal{R}_{\text{warm}}$ is persisted in the metadata generated during training.

A route collected after training remains classified as cold until a new training cycle generates updated model artifacts and metadata.

6 Prediction interpretation

For an observation i , define the residual

$$e_i = Y_i - \hat{Y}_i.$$

Thus,

$$e_i < 0$$

means that the observed price is below the estimated conditional reference, while

$$e_i > 0$$

means that the observed price is above that reference.

In an end-to-end test performed on the BSB–CNF route, the observed price was

$$Y = 232.00 \text{ USD}$$

and the model estimate was

$$\hat{Y} = 312.99 \text{ USD}.$$

Therefore,

$$e = 232.00 - 312.99 = -80.99 \text{ USD}.$$

In this case, the observation is classified as **below the model’s expected price**.

This classification does not constitute a forecast that the price will increase in the future, nor does it constitute a purchase recommendation.

7 Evaluation metrics

In addition to MAE, the system reports RMSE, R^2 , and MAPE.

$$\begin{aligned} \text{RMSE} &= \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2} \\ R^2 &= 1 - \frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \\ \text{MAPE} &= \frac{100}{n} \sum_{i=1}^n \left| \frac{Y_i - \hat{Y}_i}{Y_i} \right|. \end{aligned}$$

MAE is particularly useful in this project because it remains directly interpretable in U.S. dollars.

8 Scope limitations

The current model does not directly estimate

$$Y_{t+1},$$

nor

$$\mathbb{P}(Y_{t+1} < Y_t),$$

nor a temporal sequence

$$(Y_t, Y_{t+1}, Y_{t+2}, \dots).$$

Consequently, the system should not be characterized as a time-series model or as a future-price forecasting model.

Its current formulation is

$$\hat{Y} \approx \mathbb{E}[Y \mid X]$$

or, making the covariates explicit,

$$\hat{Y} \approx \mathbb{E}[\text{price} \mid \text{flight characteristics, search context}].$$

9 Temporal depth as future work

The main future extension is to increase the temporal depth of the dataset.

At present, the dataset contains variation in *booking lead time*, meaning that flights are searched at different values of

days_until_departure.

This is not equivalent to having a long historical trajectory for the same flight or route.

The intended temporal extension consists of repeatedly observing the same route over time, producing sequences such as

$$Y_{t-k}, Y_{t-k+1}, \dots, Y_{t-1}, Y_t.$$

With sufficient temporal depth, it would be possible to formulate a different problem, for example,

$$\mathbb{E}[Y_{t+h} \mid X_t, Y_t, Y_{t-1}, \dots, Y_{t-k}],$$

or

$$\mathbb{P}(Y_{t+h} < Y_t \mid \mathcal{H}_t),$$

where $\mathcal{H}_t$ represents the historical information available up to time t .

This extension would make it possible to investigate recent price trends, rolling statistics, route-level historical trajectories, variation relative to previous searches, and future price direction.

This problem should be treated as a separate temporal modeling task from the current expected-price estimator.

10 Summary

The system implements a supervised tabular regression model based on CatBoost to approximate

$$\mathbb{E}[Y \mid X].$$

The modeling architecture separates two regimes: warm-start for new observations from historically known routes, and cold-start for generalization to routes absent from the corresponding training set.

At inference time, the route determines which of the two model artifacts is used.

The returned value should be interpreted as an estimate of expected price conditional on the observed flight and search characteristics, rather than as a temporal forecast of future price.